Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Java/Python backend services**, REST APIs, relational data systems, and e-commerce applications; shipped tested distributed platforms handling 2M+ records and 10K+ daily requests with 99%+ reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **backend services and normalized relational schemas** using Python, FastAPI, PostgreSQL, and SQL Server for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through indexing optimization.
- Developed **asynchronous data pipelines** processing 5,000+ monthly jobs with explicit states, failure isolation, reconciliation, validation, and CI/CD releases, achieving 99%+ reliability across recurring distributed production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a **Python Flask backend** serving 150+ users and 10,000+ daily requests, implementing concurrent batch execution and fault-tolerant REST APIs while maintaining stable performance under peak production load.
- Delivered features across the **full software development lifecycle**, adding secure ingestion, validation, persistent results, usage logging, fallback handling, GitHub Actions deployments, and production troubleshooting while reducing feedback turnaround 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Troubleshot and optimized **service-oriented API integrations** using OOP, AWS Glue, S3, SQL, caching, JSON, and Parquet transformations, improving response times 30%, pipeline efficiency 68%, and storage costs 15%.
- Automated **builds, deployments, and 20+ unit/integration suites** with GitHub, Jenkins, PyTest, and Mockito, generating coverage and failure reports while reducing regressions and technical debt 20% across releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **Node.js APIs across eight microservices** for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through WebSockets and Azure SQL while reducing cross-location sync conflicts 33%.
- Implemented **automated testing and production service observability** with Jest, Selenium, Sentry, Azure Monitor, and Kibana, then packaged services through Docker and Jenkins CI/CD, cutting release time 50%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **backend APIs for order-style workflows** covering bookings, time-slot selection, reservation tracking, pricing, payments, and real-time status updates, maintaining sub-200-millisecond responses across 50,000+ records through Firebase synchronization.
- Streamlined **reliable cross-platform application delivery** by 22% through Spring Boot API and Firebase integration, implementing unit automation, version-controlled releases, and secure authentication across mobile and web experiences.

Skills

Languages: Java, C++, Python, JavaScript, TypeScript, SQL, Bash, HTML, CSS
Backend: Spring Boot, FastAPI, Flask, Node.js, Express.js, REST APIs, microservices, asynchronous processing
Data: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, Firebase, RDBMS, indexing, query tuning
Core CS: OOP, data structures, algorithms, complexity analysis, design patterns, distributed systems, concurrency
Quality & Platform: AWS S3, AWS RDS, Git, Jenkins, CI/CD, PyTest, Mockito, Jest, Selenium, Docker, observability

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered Java backend application** supporting 10+ SQL and NoSQL systems, accelerating query authoring, schema exploration, and table management through service APIs, a TypeScript interface, and Docker workflows.

CPU Scheduling Simulator Live | *C++, React.js, Algorithms*
- Implemented **six scheduling and optimization algorithms** with interactive controls, Gantt visualization, and dynamic performance metrics, testing 1,000+ process combinations while reducing average turnaround time 25% across benchmark scenarios.

Trend Tracker Live | *React.js, Strapi, AWS EC2, CNN*
- Developed an **e-commerce recommendation application** with React, Strapi, AWS EC2, and CNN-based ranking, improving recommendation accuracy from 78% to 92% while supporting more than 3M users and 200K positive retentions.